$$\vec{c}\times(\vec{a}\times\vec{b}) = (\vec{c}\cdot\vec{b})\vec{a}-(\vec{c}\cdot\vec{a})\vec{b}\,.$$

Adding the above equations and using the scalar product of two vectors is commutative, we get $\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})=\vec{0}$.

Theorem 6.10 (Lagrange's identity)

For any four vectors $\vec{a},\vec{b},\vec{c},\vec{d}$, we have $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = \begin{vmatrix} \vec{a}\cdot\vec{c} & \vec{a}\cdot\vec{d} \\ \vec{b}\cdot\vec{c} & \vec{b}\cdot\vec{d} \end{vmatrix}$.

Proof

Since dot and cross can be interchanged in a scalar product, we get

$$\begin{aligned}(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) &= \vec{a}\cdot(\vec{b}\times(\vec{c}\times\vec{d}))\\ &= \vec{a}\cdot((\vec{b}\cdot\vec{d})\vec{c}-(\vec{b}\cdot\vec{c})\vec{d}) \quad \text{(by vector triple product expansion)}\\ &= (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d})-(\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c})\\ &= \begin{vmatrix} \vec{a}\cdot\vec{c} & \vec{a}\cdot\vec{d} \\ \vec{b}\cdot\vec{c} & \vec{b}\cdot\vec{d} \end{vmatrix}\end{aligned}$$

Example 6.19

Prove that $[\vec{a}\times\vec{b},\ \vec{b}\times\vec{c},\ \vec{c}\times\vec{a}]=[\vec{a},\vec{b},\vec{c}]^2$.

Solution

Using the definition of the scalar triple product, we get

$$[\vec{a}\times\vec{b},\ \vec{b}\times\vec{c},\ \vec{c}\times\vec{a}] = (\vec{a}\times\vec{b})\cdot[(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})]. \qquad \text{... (1)}$$

By treating $(\vec{b}\times\vec{c})$ as the first vector in the vector triple product, we find

$$(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a}) = ((\vec{b}\times\vec{c})\cdot\vec{a})\vec{c}-((\vec{b}\times\vec{c})\cdot\vec{c})\vec{a} = [\vec{a},\vec{b},\vec{c}]\vec{c}\,.$$

Using this value in (1), we get

$$[\vec{a}\times\vec{b},\vec{b}\times\vec{c},\vec{c}\times\vec{a}] = (\vec{a}\times\vec{b})\cdot([\vec{a},\vec{b},\vec{c}]\vec{c}) = [\vec{a},\vec{b},\vec{c}](\vec{a}\times\vec{b})\cdot\vec{c} = [\vec{a},\vec{b},\vec{c}]^2\,.$$

Example 6.20

Prove that $(\vec{a}\cdot(\vec{b}\times\vec{c}))\vec{a} = (\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c})$.

Solution

Treating $(\vec{a}\times\vec{b})$ as the first vector on the right hand side of the given equation and using the vector triple product expansion, we get

$$(\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c}) = ((\vec{a}\times\vec{b})\cdot\vec{c})\vec{a}-((\vec{a}\times\vec{b})\cdot\vec{a})\vec{c} = (\vec{a}\cdot(\vec{b}\times\vec{c}))\vec{a}\,.$$

Example 6.21

For any four vectors $\vec{a},\ \vec{b},\ \vec{c},\ \vec{d}$, we have

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = [\vec{a},\vec{b},\vec{d}]\vec{c}-[\vec{a},\vec{b},\vec{c}]\vec{d} = [\vec{a},\vec{c},\vec{d}]\vec{b}-[\vec{b},\vec{c},\vec{d}]\vec{a}\,.$$

Solution

Taking $\vec{p}=(\vec{a}\times\vec{b})$ as a single vector and using the vector triple product expansion, we get

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = \vec{p}\times(\vec{c}\times\vec{d})$$

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